Reference Guide for Stack Tracing in Java

2026-03-28

1 Basic Java

```
public static void main(String[] args) {
    int anInt = 10;
    double aDouble = 5.8;
    boolean aBoolean = true;
    String aString = "6.3";
    anInt = 20;
}
```

- variable changes result in previous values being crossed out and new ones written in so that progression can be seen

2 Method calls

2.1 Return value

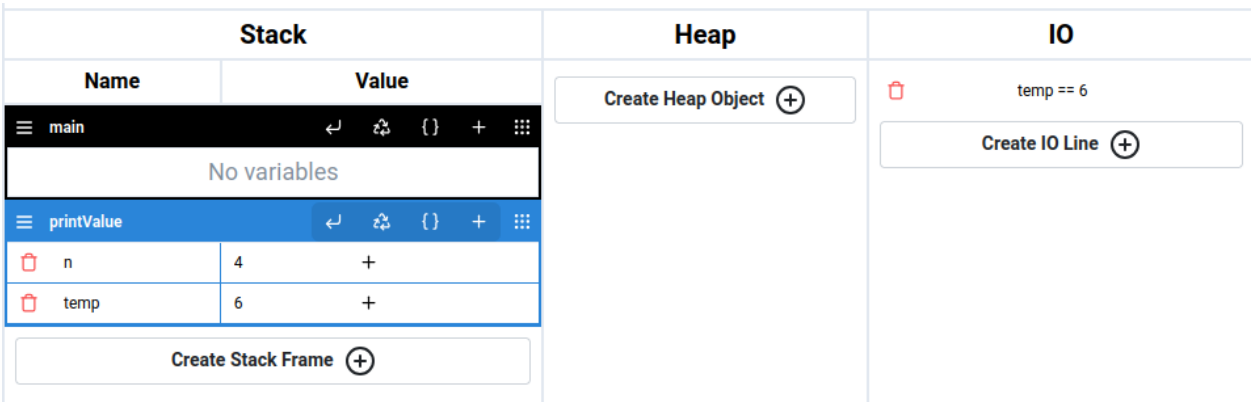
```
public static double multiplyByTwo(double input) {
    double x = input * 2;
    return x;
}

public static void main (String[] args) {
    double x = 7.0;
    double result = multiplyByTwo(x);
    result = multiplyByTwo(result);
    System.out.println(result);
    int y = 3;
}
```

- each function call is put in its own stack frame
- variables created after the function call appear further down the stack
- IO stands for input/output and is where any command line user input or outputs in terms of print statements appear

Stack		Heap	IO
Name	**Value**		28.0
main			
x	7.0		
result	~~14.0~~ 28.0		
y	3		
multiplyByTwo			
input	7.0		
x	14.0		
multiplyByTwo			
input	14.0		
x	28.0		

2.2 No return value

```
public static void printValue(int n) {
    int temp = n+2;
    System.out.println("temp == " + temp);
}

public static void main (String[] args) {
    printValue(4);
}
```



3 For loops

3.1 Basic

```
public static void main (String[] args) {
    int x = 4;
    for (int i = 1; i < 5; i++) {
        System.out.println("i == " + i);
    }
}
```

Stack		Heap	IO
Name	**Value**		i == 1
main			i == 2
x	4		i == 3
i	~~1~~ ~~2~~ ~~3~~ ~~4~~ 5		i == 4

3.2 Scoped Variable

```
public static void main (String[] args) {
    int x = 4;
    for (int i = 1; i < 5; i++) {
        int temp = i + 2;
    }
}
```

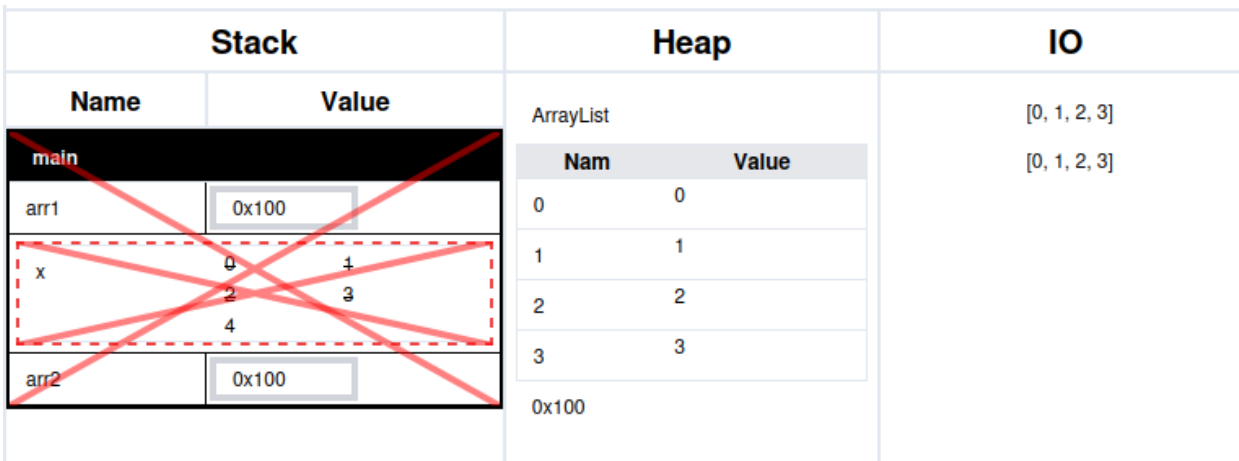
- the dashed lines indicates the variables that are scoped within the loop are crossed out after the loop completes

Stack		Heap	IO
Name	**Value**		
main			
x	4		
i	~~1~~ ~~2~~ ~~3~~ ~~4~~ 5		
temp	~~3~~ ~~4~~ ~~5~~ 6		

4 ArrayLists

```
public static void main (String[] args) {
    ArrayList<Integer> arr1 = new ArrayList<>();
    for (int x = 0; x < 4; x++) {
        arr1.add(x);
    }
    System.out.println(arr1);
    ArrayList<Integer> arr2 = arr1;
    System.out.println(arr2);
}
```

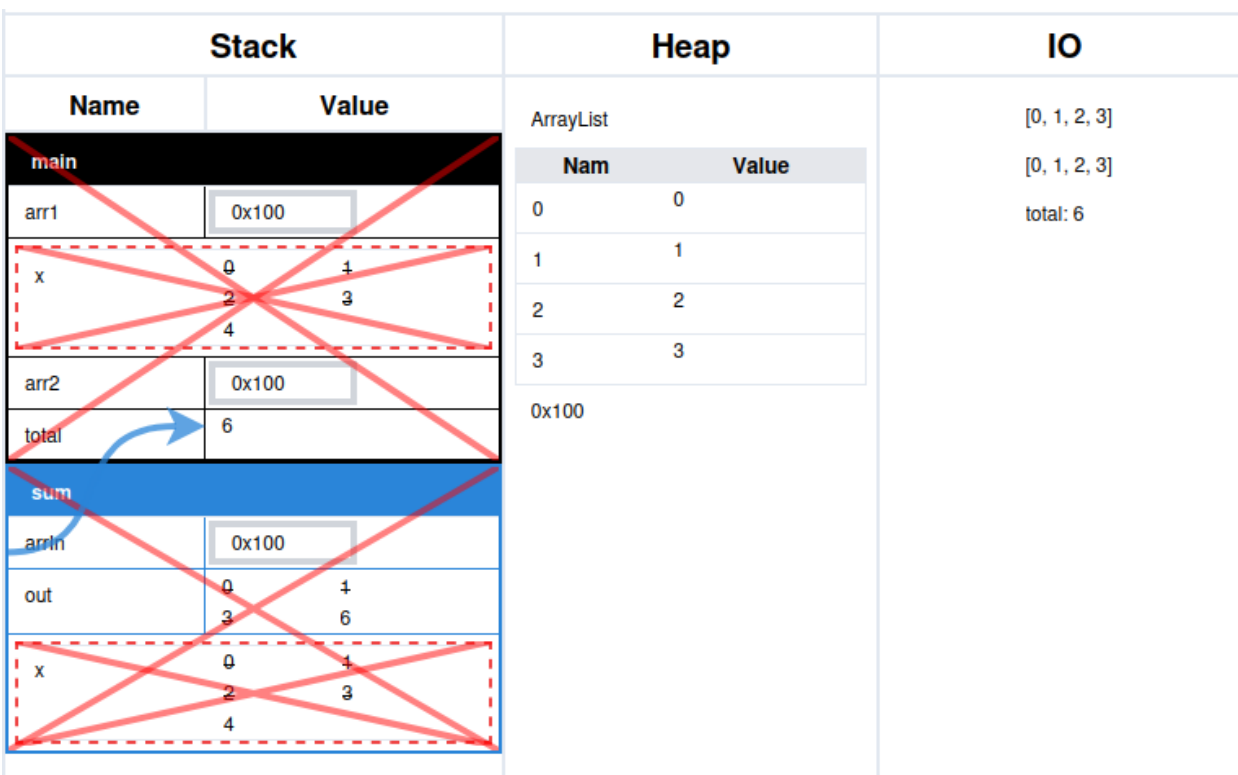
- note that the assignment statement for arr2 assigns the memory address and does not do a deep copy
- memory addresses all start with 0x to indicate that they are hexadecimal numbers.
  - heap addresses are given 3 digit numbers in the tracing tool
  - numbers effectively random. The values are not important as long it is consistent across the stack and heap



4.1 Passed to functions

```
public static int sum(ArrayList<Integer> arrIn) {
    int out = 0;
    for (int x = 0; x < arrIn.size(); x++) {
        out += arrIn.get(x);
    }
    return out;
}

public static void main (String[] args) {
    ArrayList<Integer> arr1 = new ArrayList<>();
    for (int x = 0; x < 4; x++) {
        arr1.add(x);
    }
    System.out.println(arr1);
    ArrayList<Integer> arr2 = arr1;
    System.out.println(arr2);
    int total = sum(arr1);
    System.out.println("total: " + total);
}
```

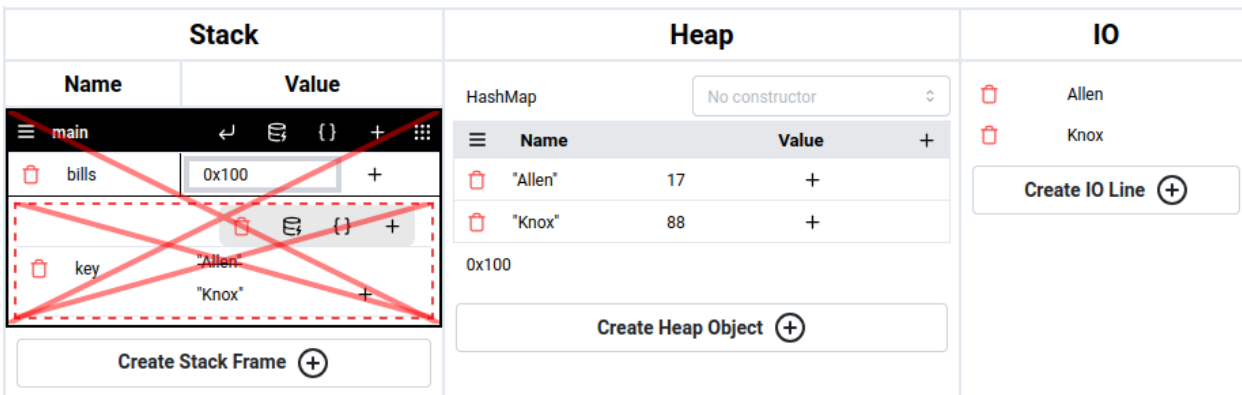
- when passing variables to functions as arguments the value on the stack associated with the variable name is the argument to the function that sets the parameter, even when that value is a reference to an object on the heap



5 HashMaps

```
public static void main(String[] args) {
    HashMap<String, Integer> bills = new HashMap<>();
    bills.put("Allen", 17);
    bills.put("Knox", 88);
    for (String key : bills.keySet()) {
        System.out.println(key);
    }
}
```

- HashMaps are like dictionaries in python
- we loop through them in a manner more similar to loops in python
- note that the loop still has scoped variables like the previous for loop



6 Recursion

6.1 Standard Recursion

```
public static int computeGeometricSum(int n) {
    if (n > 0) {
        int result = computeGeometricSum(n - 1);
        result += n;
        return result;
    } else {
        return 0;
    }
}

public static void main (String[] args) {
    int result = computeGeometricSum(3);
}
```

- each new call of computeGeometricSum is performed in a new color
  - returned values are kept in the color of the method that returns it

6.2 Tail Recursion

```
public static int computeGeometricSumTail(int n, int total) {
    if (n > 0) {
        return computeGeometricSumTail(n - 1, total + n);
    } else {
        return total;
    }
}

public static int cGSTHelper(int n) {
    return computeGeometricSumTail(n, 0);
}

public static void main (String[] args) {
    int result = cGSTHelper(3);
}
```

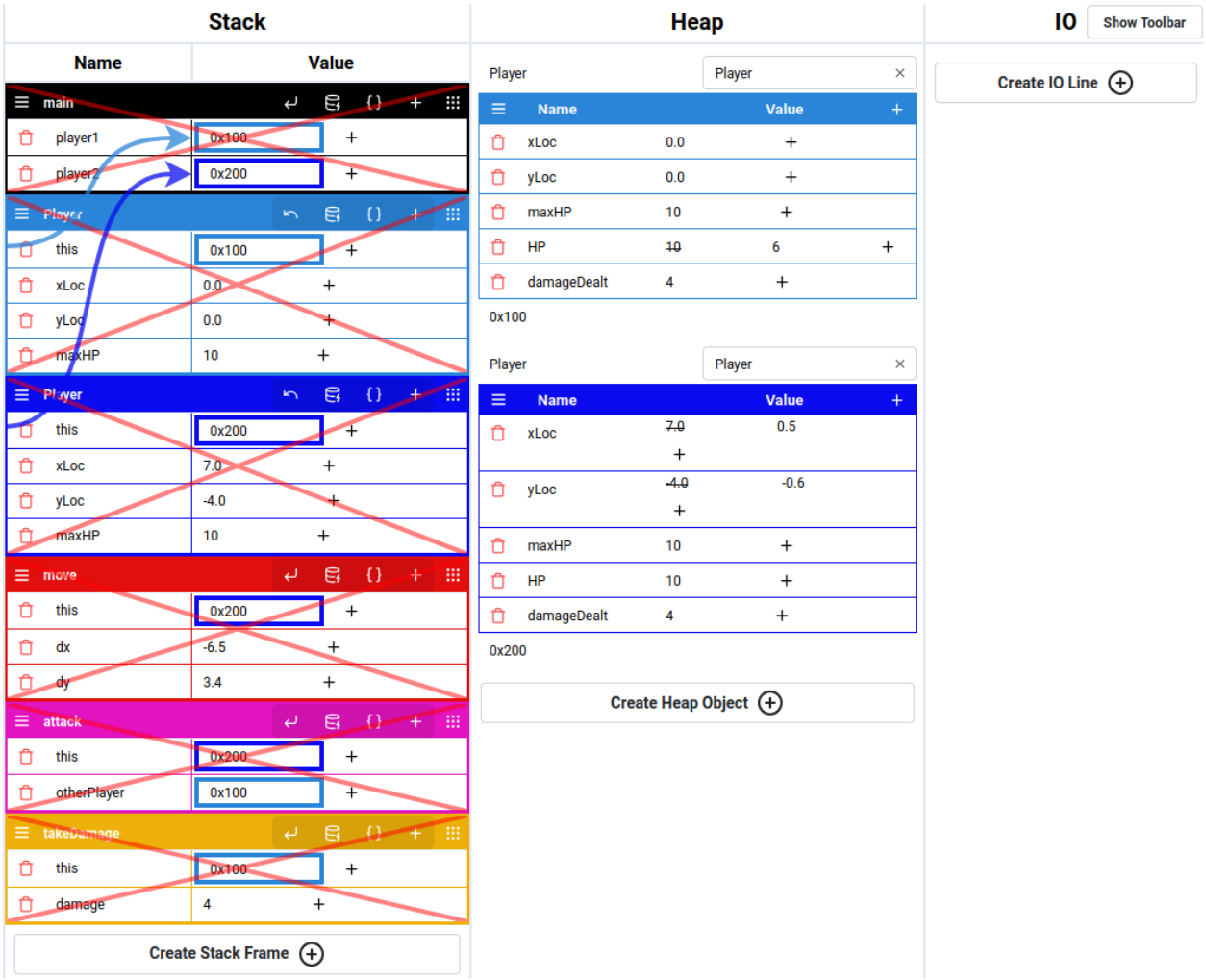
- Note that all the return arrows point to the original return variable when using tail recursion. It is also acceptable to omit the return arrows entirely since the recursive calls do not return direectly to a variable

(Solution on next page)

Stack		Heap	IO
Name	**Value**		
main			
result	6		
cGSTHeader			
n	3		
computeGeometric			
n	3		
total	0		
computeGeometric			
n	2		
total	3		
computeGeometric			
n	1		
total	5		
computeGeometric			
n	0		
total	6		

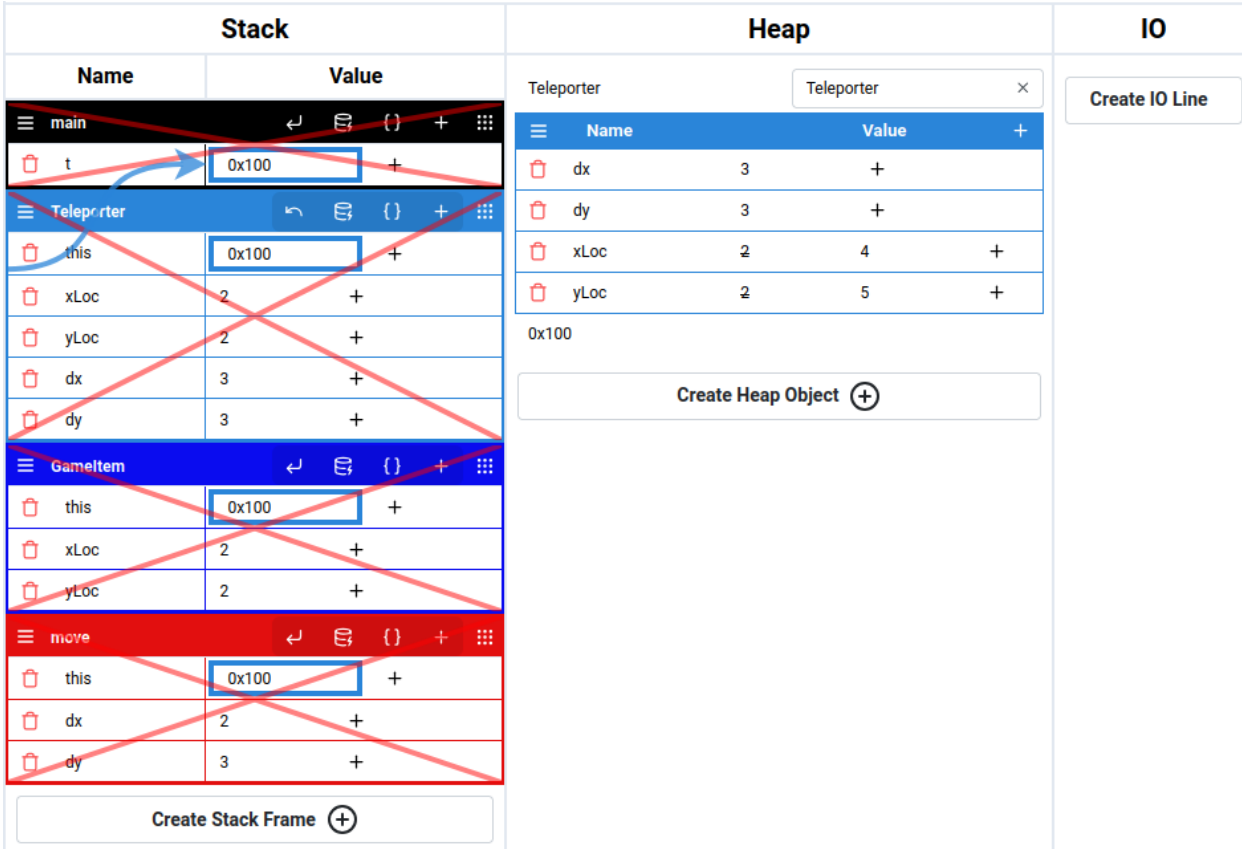
7 Classes

```
public class Player {
    private double xLoc;
    private double yLoc;
    private int maxHP;
    private int HP;
    private int damageDealt;

    public Player(double xLoc, double yLoc, int maxHP) {
        this.xLoc = xLoc;
        this.yLoc = yLoc;
        this.maxHP = maxHP;
        this.HP = maxHP;
        this.damageDealt = 4;
    }
    public int getHP() {
        return this.HP;
    }
    public void takeDamage(int damage) {
        this.HP -= damage;
    }
    public void attack(Player otherPlayer) {
        otherPlayer.takeDamage(this.damageDealt);
    }
    public void move(double dx, double dy) {
        this.xLoc += dx;
        this.yLoc += dy;
    }

    public static void main(String[] args) {
        Player player1 = new Player(0.0, 0.0, 10);
        Player player2 = new Player(7.0, -4.0, 10);
        player2.move(-6.5, 3.4);
        player2.attack(player1);
    }
}
```


Stack
Name
Value
main
player1
0x100
player2
0x200
Player
this
0x100
xLoc
0.0
yLoc
0.0
maxHP
10
Player
this
0x200
xLoc
7.0
yLoc
-4.0
maxHP
10
move
this
0x200
dx
-6.5
dy
3.4
attack
this
0x200
otherPlayer
0x100
takeDamage
this
0x100
damage
4
Create Stack Frame
Heap
Player
Player
Name
Value
xLoc
0.0
yLoc
0.0
maxHP
10
HP
10
6
damageDealt
4
0x100
Player
Player
Name
Value
xLoc
7.0
0.5
yLoc
-4.0
-0.6
maxHP
10
HP
10
damageDealt
4
0x200
Create Heap Object
IO
Show Toolbar
Create IO Line

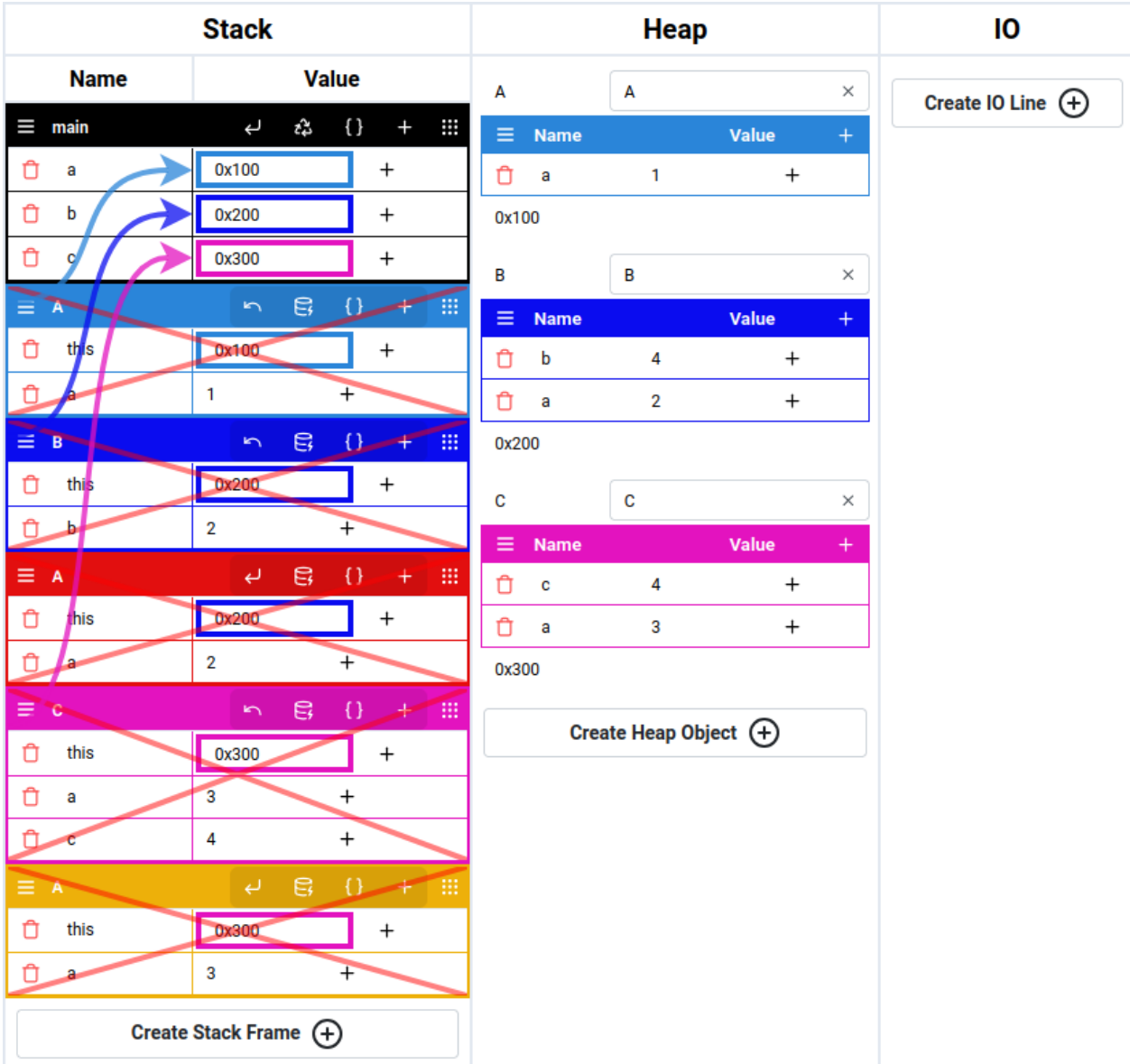
8 Inheritance

```
public class GameItem {
    private double xLoc;
    private double yLoc;

    public GameItem(double xLoc, double yLoc) {
        this.xLoc = xLoc;
        this.yLoc = yLoc;
    }
    public void move(double dx, double dy) {
        this.xLoc += dx;
        this.yLoc += dy;
    }
}

public class Teleporter extends GameItem {
    private double dx;
    private double dy;

    public Teleporter(double xLoc, double yLoc, double dx, double dy) {
        super(xLoc, yLoc);
        this.dx = dx;
        this.dy = dy;
    }

    public static void main(String[] args) {
        Teleporter t = new Teleporter(2, 2, 3, 3);
        t.move(2, 3);
    }
}
```


Stack
Heap
IO
Name
Value
main
t
0x100
Teleporter
this
0x100
xLoc
2
yLoc
2
dx
3
dy
3
GameItem
this
0x100
xLoc
2
yLoc
2
move
this
0x100
dx
2
dy
3
Create Stack Frame
Teleporter
Teleporter
Name
Value
dx
3
3
dy
3
3
xLoc
2
4
yLoc
2
5
0x100
Create Heap Object
Create IO Line

9 Polymorphism

```
public class A {
    protected int a;

    public A(int a) {
        this.a = a;
    }
}

public class B extends A{
    private int b;

    public B(int b) {
        super(b);
        this.b = b*2;
    }
}

public class C extends A{
    private int c;

    public C(int a, int c) {
        super(a);
        this.c = c;
    }
}

public class RunABC {
    public static void main(String[] args) {
        A a = new A(1);
        A b = new B(2);
        A c = new C(3, 4);
    }
}
```


Stack
Heap
IO
Name
Value
main
a
0x100
b
0x200
c
0x300
A
this
0x100
a
1
B
this
0x200
b
2
A
this
0x200
a
2
C
this
0x300
a
3
c
4
A
this
0x300
a
3
Create Stack Frame
A
A
Name
Value
a
1
0x100
B
B
Name
Value
b
4
a
2
0x200
C
C
Name
Value
c
4
a
3
0x300
Create Heap Object
Create IO Line